

MCP: Connecting Claude to live data

The full reference. For the one-page version, see *mcp-for-faculty-overview*.

WHAT MCP IS (START HERE)

Claude has a blind spot: it can only work with its training data (cutoff date) and what you paste in. It can't go look something up. **MCP** — Model Context Protocol — changes that. It connects Claude to external services (databases, search engines, APIs) so it can query them in real time. Mental model: **MCP servers are like browser extensions for Claude**. They don't change what Claude *is* — they change what Claude *can do*.

- 1

You don't have to build anything to use one. If an MCP is set up in your project (like Semantic Scholar in this workshop), you just talk to Claude. *"Find papers about X"* works.
- 2

An MCP is not magic. It's a small program (~400 lines of Python) that translates Claude's requests into API calls and formats results.
- 3

The results are really real. When Claude uses an MCP to search Semantic Scholar, every paper actually exists, with real citation counts and real DOIs — fundamentally different from recall from memory, where it might hallucinate.

WHAT'S AN API?

A structured way for one program to request data from another. When you search Semantic Scholar in your browser, you're using their website — behind it is an API designed for programs to call directly. **An MCP server makes those same structured requests on Claude's behalf**. Most major academic services have APIs: Semantic Scholar, PubMed, CrossRef, Canvas, Zotero. (Google Scholar notably doesn't.) Whether a service has one — and how well documented it is — determines whether an MCP can connect.

WHAT IT IS NOT

Not an API itself — a standard way to *connect* to APIs. Not exclusive to Anthropic; the spec is open. Not required to use Claude — most users never touch it.

WHY THIS MATTERS FOR FACULTY WORK

FOR RESEARCH

Faculty already use specialized databases (PubMed, JSTOR, Semantic Scholar, Web of Science). MCP lets Claude query these directly.

- **Without MCP:** *"What papers exist on grade inflation?"* Possibly outdated or wrong training data; may invent plausible-sounding citations.
- **With MCP:** *"Search Semantic Scholar for papers on grade inflation."* Live query, real results, filterable by year, field, citation count.

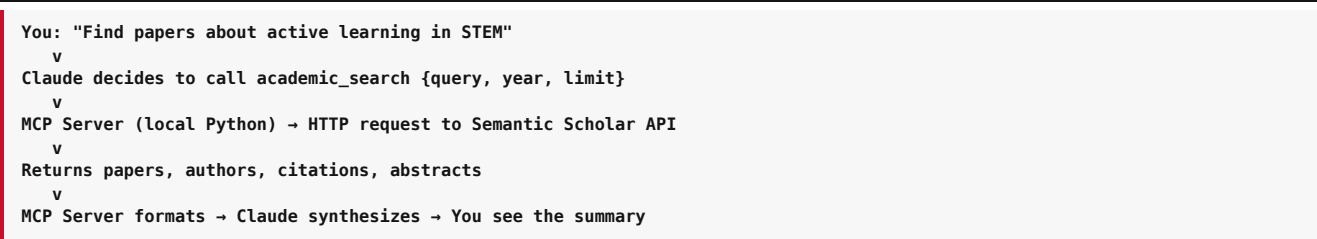
FOR TEACHING

Reading lists grounded in real sources (library catalog or discipline database). **Student-facing tools** — an MCP connected to Canvas could help students navigate course materials, deadlines, or feedback within institutional guardrails.

WHY UNDERSTAND THE MECHANISM (EVEN IF YOU NEVER BUILD ONE)

You can **evaluate AI tools more critically** when a vendor says "our AI is connected to your data." You can **make better requests**: *"Could we build an MCP that connects Claude to our IRB database?"* beats *"can AI do IRB stuff?"* And you understand the **limits** — an MCP is only as good as the API it wraps.

HOW IT WORKS



Using, troubleshooting, building

A sequence to try right now in Claude Code, plus the common questions and the three paths if you want to build one.

HOW CLAUDE DISCOVERS MCP TOOLS, WHERE TO REGISTER THEM

When Claude Code starts, it reads *.mcp.json* in your project directory. The file lists each MCP server: program to run, arguments, env vars (like API keys). Claude Code launches each server and discovers its tools.

Claude Code (terminal): *.mcp.json* in the project (project-scoped) or *~/claude/settings.json* (global, via */config* or *claude mcp add*). **Desktop app**: macOS — *~/Library/Application Support/Claude/claude_desktop_config.json*; Windows — *%APPDATA%\Claude\claude_desktop_config.json*. Practical difference: Claude Code supports project-scoped MCPs (collaborators get them automatically); Desktop is global-only.

API KEYS

A credential that identifies you to the external service — like a library card. Semantic Scholar: **without a key**, ~1 request/5s (you'll hit rate limits constantly). **With a free key**, ~10/s. Get one at *semanticsscholar.org/product/api* (30 seconds, no credit card). The key goes in *.mcp.json*, not the code — so code can be shared without sharing credentials.

TRY IT YOURSELF

A sequence to run right now using the Semantic Scholar MCP already set up:

- 1

Search. *"Find recent papers about [your topic]"* — Claude narrates: *"I'm calling academic_search with..."*
- 2

Go deeper. *"Tell me more about the [Author Year] paper — who cites it?"* Calls *academic_get_paper*.
- 3

Discover. *"Find papers related to this one."* Calls *academic_recommend* (citation-graph patterns).
- 4

Save. *"Search for papers on [topic] and save the results."* A markdown file appears in *output/*.

Pattern: ask in English → Claude translates to a tool call → MCP makes the API request → results come back. By the third or fourth query, it feels natural.

TROUBLESHOOTING

- **"Rate limited":** API key isn't loaded. Check *.mcp.json* and restart Claude Code.
- **"Paper not found":** DOIs need the *DOI*: prefix (e.g., *DOI:10.1038/...*).
- **Off-topic results:** Add filters: *"...in the Education field from 2020 onward."*
- **Tools not appearing:** Run *bash _mcp/academic_search/setup.sh* and restart.

COMMON QUESTIONS

- **"Can someone build one for [my database]?"** If it has a web API, yes. A well-documented API takes 1–2 hours. Institutionally authenticated systems take longer — talk to the Bok Center.
- **"Is my data safe?"** The MCP runs locally; API calls go from your machine to the external service. Anthropic sees conversation text (including tool results) under standard Claude terms. *As of June 2026, Claude Code conversations are not used to train models — check current terms.*
- **"How is this different from just googling?"** Claude can make multiple searches, filter, and synthesize without managing tabs. Results are structured data (citations, DOIs, authors). Claude can chain tools in one conversation.

IF YOU WANT TO BUILD ONE (THREE PATHS)

- **Let Claude build it.** Type */mcp-builder* in Claude Code, paste the API docs, and Claude proposes tools, generates the server code and setup files. Expect to iterate on error handling and formatting.
- **Copy and modify an existing one.** The *academic_search* MCP is a good template for any "search an external database" use case. See the *MCP Registry* and Anthropic's *modelcontextprotocol/servers* repo.
- **From scratch.** Walkthrough in *_mcp/academic_search/how-to-build-an-mcp.md* documents every design decision.

First step in any path:

get the API documentation. The */mcp-builder* can only generate correct code from accurate information.

Links: *modelcontextprotocol.io* · *docs.anthropic.com/en/docs/agents-and-tools/mcp* · *api.semanticscholar.org/api-docs* · *gofastmcp.com*

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